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THE HARRIET LANE HANDBOOK

THE JOHNS HOPKINS HOSPITAL

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A MANUAL FOR PEDIATRIC HOUSE OFFICERS

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24

THE **HARRIET LANE** HANDBOOK

THE HARRIET LANE HOUSE STAFF AT
THE CHARLOTTE R. BLOOMBERG CHILDREN'S
CENTER OF THE JOHNS HOPKINS HOSPITAL

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PEDIATRIC PARAMETERS AND EQUIPMENT

	Premie	Newborn	6 mo	1 yr	2-3 yr	4-6 yr	7-10 yr	11-15 yr	>16 yr
WT (kg)	1-3	2-4	6-8	10	12-16	20-25	25-35	40-50	>50
BAG VALVE MASK	Infant	Infant	Small child	Small child	Child	Child	Child/ small adult	Adult	Adult
NASAL AIRWAY (Fr)	12	12	14-16	14-16	14-18	14-18	16-20	18-22	22-36
ORAL AIRWAY	Infant 50 mm	Small 60 mm	Small 60 mm	Small 60 mm	Small 70 mm	Small 70-80 mm	Med 80-90 mm	Med 90 mm	Med 90 mm
BLADE	MIL 0	MIL 0	MIL 1	MIL 1, MAC 2	MIL 1, MAC 2	MIL 2, MAC 2	MIL 2, MAC 2	MIL 2, MAC 3	MIL 2, MAC 3
ETT	2.5-3.0	3.0-3.5	3.5	4.0	4.0-4.5	4.5-5.0	5.5-6.0	6.0-6.5	7.0-8.0
LMA	1	1	1.5	2	2	2.5	2.5-3	3	4
GLIDESCOPE	1	1 or 2	2	2	3	3	3	3 or 4	3 or 4
IV CATH (ga)	22-24	22-24	20-24	20-24	18-22	18-22	18-22	18-20	18-20
CVL (Fr)	3	3-4	4	4-5	4-5	5	5	7	7
NGT/OGT (Fr)	5	5-8	8	10	10-12	12-14	12-14	14-18	14-18
CHEST TUBE (Fr)	10-12	10-12	12-18	16-20	16-24	20-28	20-32	28-38	28-42
FOLEY (Fr)	6	8	8	8	8	8	8	10	12

ESTIMATED BLOOD PRESSURE BY AGE

Measurement	50th %	5th %
Systolic BP [Diastolic BP is usually 2/3 systolic BP]	90 + (age × 2)	60 (neonate); 70 (1 mo-1 yr) 70 + (age × 2) (for 2-10 yr) 90 (>10 yr)
MAP	55 + (age × 1.5)	40 + (age × 1.5)

NORMAL VITAL SIGNS BY AGE

Age	Heart Rate (beats/min)	Blood Pressure (mmHg)	Respiratory Rate (breaths/min)
Premie	120-170	55-75/35-45 (gestational age approximates normal MAP)	40-70
0-3 mo	100-150	65-85/45-55	35-55
3-6 mo	90-120	70-90/50-65	30-45
6-12 mo	80-120	80-100/55-65	25-40
1-3 yr	70-110	90-105/55-70	20-30
3-6 yr	65-110	95-110/60-75	20-25
6-12 yr	60-95	100-120/60-75	14-22
>12 yr	55-85	110-135/65-85	12-18

ENDOTRACHEAL TUBE FORMULAS

Uncuffed ETT size: age (years)/4 + 4; Cuffed ETT size: age (years)/4 + 3
ETT depth (from lip to mid-trachea): ETT internal diameter (size) × 3

GLASGOW COMA SCALE

Activity	Score	Child/Adult	Score	Infant
Eye opening	4	Spontaneous	4	Spontaneous
	3	To speech	3	To speech/sound
	2	To pain	2	To painful stimuli
	1	None	1	None
Verbal	5	Oriented	5	Coos/babbles
	4	Confused	4	Irritable cry
	3	Inappropriate	3	Cries to pain
	2	Incomprehensible	2	Moans to pain
	1	None	1	None
Motor	6	Obeys commands	6	Normal spontaneous movement
	5	Localizes to pain	5	Withdraws to touch
	4	Withdraws to pain	4	Withdraws to pain
	3	Abnormal flexion	3	Abnormal flexion (decorticate)
	2	Abnormal extension	2	Abnormal extension (decerebrate)
	1	None (flaccid)	1	None (flaccid)

Adapted from *Hunt EA, Nelson-McMillan K, Levin A, O'Brien C, McNamara L, Hutchins L. The Johns Hopkins Children's Center Kids Kard, 2021.*

RESUSCITATION MEDICATIONS

Adenosine Supraventricular tachycardia	**0.1 mg/kg IV/IO RAPID BOLUS (over 1-2 sec), Flush with 10 mL normal saline** May repeat at 0.2 mg/kg IV/IO, then 0.3 mg/kg IV/IO after 2 min Max first dose 6 mg, max subsequent dose 12 mg Administer using a 3-way stopcock attached to a 10 mL NS flush or can consider administration combined via a single syringe for adult sized patients. Prolonged asystole may occur following administration
Amiodarone Ventricular tachycardia Ventricular fibrillation	**5 mg/kg IV/IO** No Pulse: Push Undiluted Pulse: Dilute and give over 20-60 minutes Max first dose 300 mg, max subsequent dose 150 mg Max total dose 15 mg/kg/24 hours OR 2200 mg/24 hours Strongly consider pretreating with IV calcium in patients with a pulse to prevent hypotension. Monitor for hypotension after administration
Atropine Bradycardia (increased vagal tone) Primary AV block	**0.02 mg/kg IV/IO/IM, 0.04–0.06 mg/kg ETT** Max single dose 0.5 mg Repeat once in 5 minutes if needed to max total dose 1 mg. Max total dose 3 mg used in adults. 1 mg/mL concentration recommended for IM or ETT use
Calcium chloride (10% elemental calcium) Hypocalcemia Hyperkalemia Hypermagnesemia Calcium channel blocker overdose	**20 mg/kg IV/IO** Max dose 1,000 mg May administer peripherally during an arrest
Calcium gluconate (10% elemental calcium) Hypocalcemia Hyperkalemia Hypermagnesemia Calcium channel blocker overdose	**60 mg/kg IV/IO** Max dose 3,000 mg
Dextrose Hypoglycemia	**Weight-Based Dosing: 0.5–1 gram/kg** **Volume-Based Dosing ("Rule of 50"):** **<5 kg: 10% dextrose 5-10 mL/kg IV/IO** **5-44 kg: 25% dextrose 2-4 mL/kg IV/IO** **≥45 kg: 50% dextrose 1-2 mL/kg IV/IO** **Max single dose 25 grams = 50 mL**
Epinephrine Pulseless arrest Bradycardia (symptomatic) Anaphylaxis	**0.01 mg/kg IV/IO (0.1 mg/mL) every 3–5 min (max single dose 1 mg)** **0.1 mg/kg ETT (1 mg/mL) every 3–5 min (max single dose 2.5 mg)** **Anaphylaxis:** **0.01 mg/kg IM (1 mg/mL) in thigh every 5-15 min PRN; max single dose 0.5 mg** **Standardized/Autoinjector:** **<7.5 kg: no autoinjector, see above** **7.5 to <15 kg: 0.1 mg IM** **15 to <30 kg: 0.15 mg IM** **≥30 kg: 0.3 mg IM**
Hydrocortisone Adrenal Crisis/Insufficiency	**2 mg/kg IV/IM/IO** Max dose 100 mg
Insulin (Regular or Aspart) Hyperkalemia	**0.1 units/kg IV/IO with 0.5 gram/kg of dextrose** Max dose 5 units Doses less than 5 units require dilution. Consider using a 1 unit/mL dilution.
Lidocaine Antiarrhythmic	**1 mg/kg IV/IO, 2-3 mg/kg ETT** Max single dose 100 mg May repeat every 5 min to max total dose 3 mg/kg

Magnesium sulfate Torsades de pointes Hypomagnesemia	**50 mg/kg IV/IO** No Pulse: Dilution recommended; infusion rate dependent on indication Pulse: Dilute and give over 20-60 minutes Max single dose 2,000 mg Monitor for hypotension/bradycardia
Naloxone Opioid overdose Coma	**Respiratory Depression: 0.001-0.005 mg/kg/dose IV/IO/IM/SUBQ (max 0.1 mg first dose, may titrate to effect)** **Full Reversal/Arrest Dose: 0.1 mg/kg IV/IO/IM/SUBQ (max dose 2 mg)** ETT dose 2–3 times IV dose, IN dose up to 8 mg. May give every 2 min PRN
Sodium Bicarbonate (8.4% = 1 mEq/mL) Administer only with clear indication: Metabolic acidosis Hyperkalemia Tricyclic antidepressant overdose	**1 mEq/kg IV/IO** ≥10kg: Administer undiluted 1 mEq/mL product <10kg: Dilute 8.4% sodium bicarbonate 1:1 with sterile water to a final concentration of 4.2% = 0.5 mEq/mL Hyperkalemia: Max single dose 50 mEq

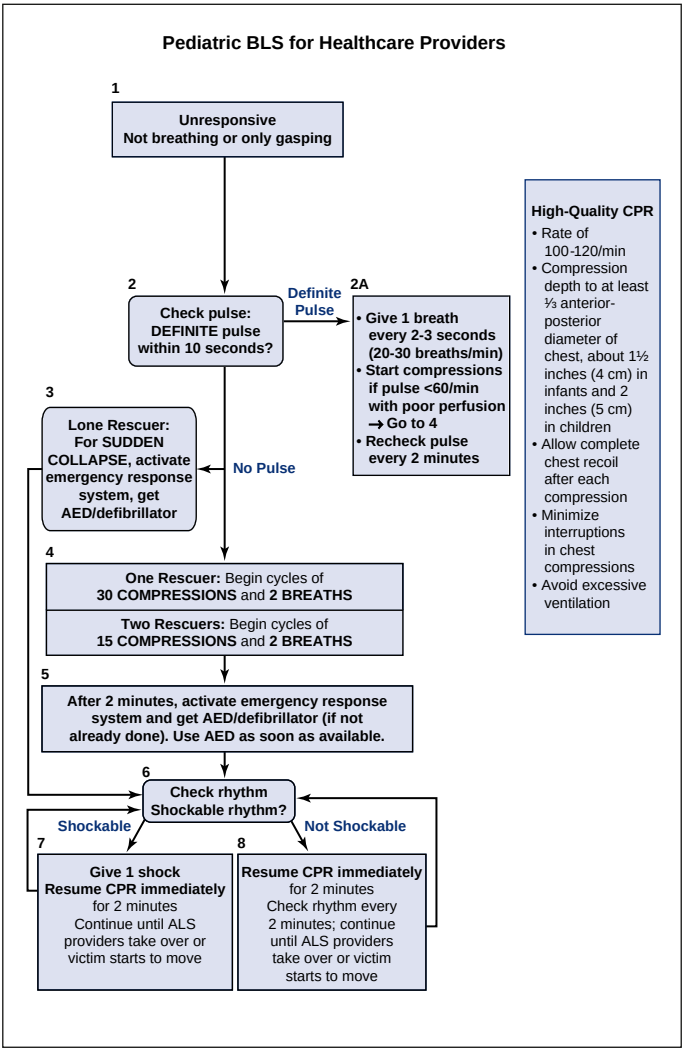
ETT Meds (NAVEL: naloxone, atropine, vasopressin, epinephrine, lidocaine)—dilute meds to 5 mL with NS, follow with positive-pressure ventilation.

Adapted from *Hunt EA, Nelson-McMillan K, Levin A, O'Brien C, McNamara L, Hutchins L. The Johns Hopkins Children's Center Kids Kard, 2021; and the American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care, 2025.*

IV INFUSIONS*

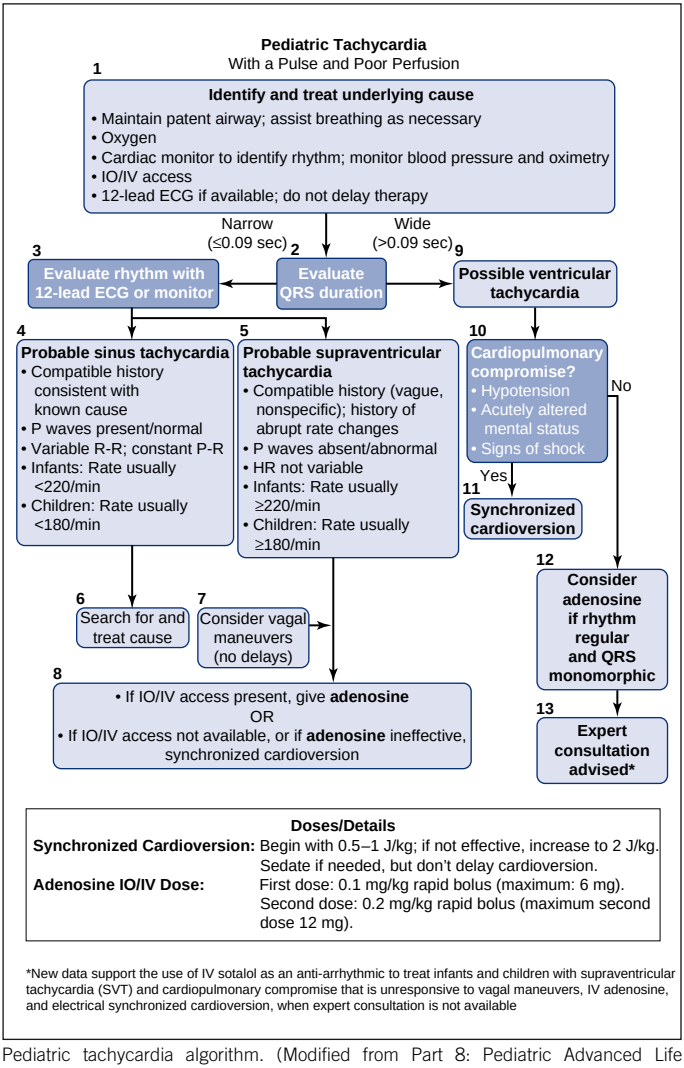
Medication	Dose (mCg/kg/min)	Dilution in 100 mL in a Compatible IV Fluid	IV Infusion Rate
Alprostadil (prostaglandin E_1)	0.05–0.1	0.3 mg/kg	1 mL/hr = 0.05 mCg/kg/min
Amiodarone	5–15	6 mg/kg	1 mL/hr = 1 mCg/kg/min
DOBUTamine	2–20	6 mg/kg	1 mL/hr = 1 mCg/kg/min
DOPamine	5–20	6 mg/kg	1 mL/hr = 1 mCg/kg/min
EPINEPHrine	0.05-2	0.6 mg/kg	1 mL/hr = 0.1 mCg/kg/min
Lidocaine, post resuscitation	20–50	6 mg/kg	1 mL/hr = 1 mCg/kg/min
Phenylephrine	0.1–2, up to 5 in severe circumstances	0.3 mg/kg	1 mL/hr = 0.05 mCg/kg/min
Terbutaline	0.1–4 (up to 10 has been used)	0.6 mg/kg	1 mL/hr = 0.1 mCg/kg/min
Vasopressin (pressor)	0.17–8 milliunits/kg/min	6 units/kg	1 mL/hr = 1 milliunit/kg/min

* Standardized concentrations and use of smart infusion pumps are recommended when available. For additional information, see Larsen GY, Park HB et. al. Standard drug concentrations and smart-pump technology reduce continuous-medication-infusion errors in pediatric patients. *Pediatrics*. 2005;116(1):e21-e25. Special thanks to LeAnn McNamara, PharmD for her expert guidance with IV infusion and resuscitation medication guidelines.

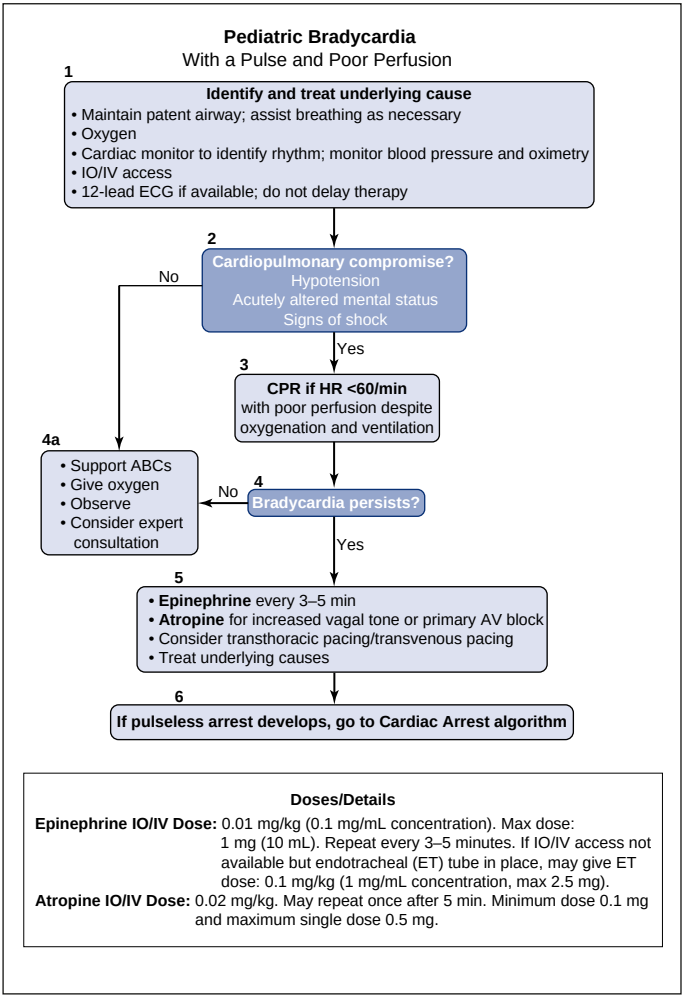


Pediatric BLS health care providers algorithm. (Modified from Part 6: Pediatric Basic Life Support: 2025 American Heart Association and American Academy of Pediatrics Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Joyner BL Jr, Dewan M, Bavare A, et al. *Circulation* Volume 152, Number 16_suppl_2. https://doi.org/10.1161/CIR.0000000000001370)

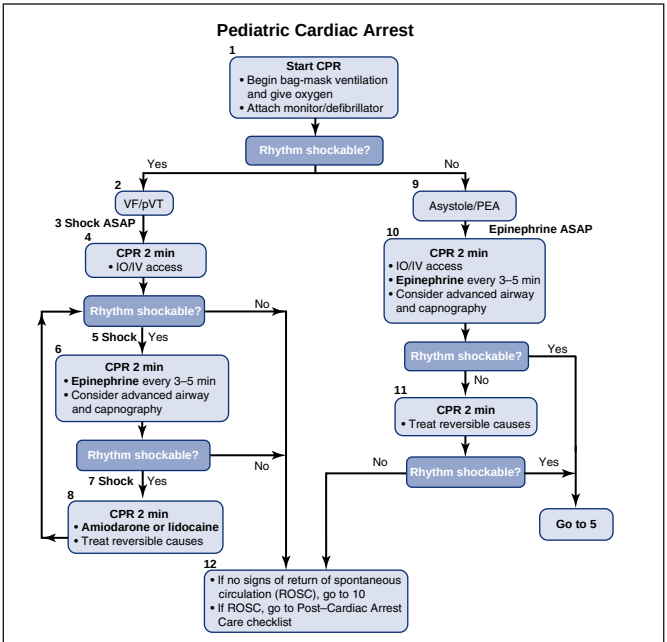
"Special thanks to Nicole Shilkofski MD, MEd for her expert guidance with the pediatric resuscitation algorithms."



Pediatric tachycardia algorithm. (Modified from Part 8: Pediatric Advanced Life Support: 2025 American Heart Association and American Academy of Pediatrics Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Lasa J, Dhillon GS, Duff JP, et al. *Circulation* Volume 152, Number 16_suppl_2. https://doi.org/10.1161/CIR.0000000000001368)



Pediatric bradycardia algorithm. (Modified from Part 8: Pediatric Advanced Life Support: 2025 American Heart Association and American Academy of Pediatrics Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Lasa J, Dhillon GS, Duff JP, et al. *Circulation* Volume 152, Number 16_suppl_2. https://doi.org/10.1161/CIR.0000000000001368)



High Quality CPR
- Push hard (≥⅓ of chest depth) and fast (100–120/min) and allow complete chest recoil
- Minimize interruptions in compressions
- Avoid excessive ventilation
- Monitor ETCO2. If possible, monitor invasive diastolic BP
- Rotate compressor every 2 minutes or sooner if fatigued
- If no advanced airway, compression to ventilation ratio: 15:2- 2 rescuers (pre-puberty)
  30:2 - 2 rescuers (post puberty)
  30:2 - 1 rescuer (any age)
- If advanced airway, provide continuous compressions and give a breath every 2-3 seconds

Shock Energy for Defibrillation
First shock 2 J/kg, second shock 4 J/kg, subsequent shocks ≥4 J/kg, maximum 10 J/kg or adult dose.

Drug Therapy
- **Epinephrine IO/IV Dose:** 0.01 mg/kg (0.1 mg/mL concentration). Max dose: 1 mg (10 mL).
  Repeat every 3–5 minutes. If no IO/IV access, may give endotracheal dose: 0.1 mg/kg (1 mg/mL concentration, max 2.5 mg)
- **Amiodarone IO/IV Dose:** 5 mg/kg bolus during cardiac arrest (max 150mg subsequent doses). Max dose: 300 mg. May repeat up to 3 total doses
- **Lidocaine IV/IO dose:** 1 mg/kg

Advanced Airway
- Endotracheal intubation or supraglottic advanced airway
- ETCO2 to confirm and monitor ET tube placement

Reversible Causes
- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypoglycemia
- Hypokalemia/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

Pediatric cardiac arrest algorithm. (Reprinted with permission *Circulation* 2025;152: S479-S537 ©2025 American Heart Association, Inc.)